

Experiment 3

Aim: Perform text tokenization using NLTK.

Theory:

Text Tokenization is a fundamental preprocessing step in Natural Language Processing (NLP) that involves breaking a given text into smaller units called *tokens*. These tokens can be sentences, words, or even characters. Tokenization helps transform unstructured text into a structured format that can be easily analyzed by machines.

The **Natural Language Toolkit (NLTK)** is a popular Python library used for NLP tasks. It provides efficient and easy-to-use tokenizers that handle punctuation, special characters, and language-specific rules.

Types of Tokenization

1. **Sentence Tokenization:**
Sentence tokenization divides a paragraph or document into individual sentences. NLTK uses pre-trained models to accurately detect sentence boundaries.
2. **Word Tokenization:**
Word tokenization splits sentences into individual words and symbols. It separates punctuation from words and preserves meaningful tokens for further analysis.

NLTK Tokenizers

NLTK provides built-in functions for tokenization:

- `sent_tokenize()` for sentence tokenization
- `word_tokenize()` for word tokenization

These tokenizers use linguistic rules and probabilistic models to ensure accurate segmentation of text.

Applications of Tokenization

- Text preprocessing for NLP pipelines
- Information retrieval systems
- Sentiment analysis
- Machine translation
- Text classification and clustering

Source Code and Outputs:

```
[1]
✓ 9s
!pip install nltk

Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)

[2]
✓ 7s
import nltk
nltk.download('punkt')
nltk.download('punkt_tab')

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
True
```

```
[3]
✓ 0s
import nltk
from nltk.tokenize import sent_tokenize, word_tokenize

text = """The sky turned a soft orange as the sun began to set behind the mountains.
        Birds chirped their final songs of the day, and a light breeze rustled through the trees.
        On a quiet evening like this, it is easy to forget the hustle and bustle of city life,
        finding peace in the simple, passing moments of nature.
        """

sentences = sent_tokenize(text)
print("Sentence Tokenization:")
print(sentences)
print()

words = word_tokenize(text)
print("Word Tokenization:")
print(words)
```

```
Sentence Tokenization:
['The sky turned a soft orange as the sun began to set behind the mountains.', 'Birds chirped their final songs of the day, and a light breeze rustled through the trees. On a quiet evening like this, it is easy to forget the hustle and bustle of city life, finding peace in the simple, passing moments of nature.']

Word Tokenization:
['The', 'sky', 'turned', 'a', 'soft', 'orange', 'as', 'the', 'sun', 'began', 'to', 'set', 'behind', 'the', 'mountains', '.', 'Birds', 'chirped', 'their', 'final', 'songs', 'of', 'the', 'day', 'and', 'a', 'light', 'breeze', 'rustled', 'through', 'the', 'trees', 'On', 'a', 'quiet', 'evening', 'like', 'this', 'it', 'is', 'easy', 'to', 'forget', 'the', 'hustle', 'and', 'bustle', 'of', 'city', 'life', 'finding', 'peace', 'in', 'the', 'simple', 'passing', 'moments', 'of', 'nature']
```

Learning Outcomes:

I am able to perform text tokenization using NLTK by breaking down raw text into sentences and words, enabling structured preprocessing and forming the foundation for further Natural Language Processing operations.